

MEASURING THE STRENGTH OF PARTIAL ENCRYPTION SCHEMES

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ABSTRACT

Partial encryption (PE) of compressed multimedia can greatly reduce the computational complexity by encrypting only a fraction of the data bits. It can also easily provide users with low-quality versions, while maintaining the high-quality version inaccessible to unauthorized users. However, it is necessary to realistically evaluate its security strength. Some of the cryptanalysis done for these techniques ignored important characteristics of the multimedia files, and used overly optimistic assumptions. We demonstrate potential weaknesses of such techniques studying attacks that exploit the information provided by non-encrypted bits, and the availability of side information (e.g., from analog signals). We show that a more useful measure of encryption strength is the complexity to reduce distortion, instead of recovering the encryption key. We consider attacks on PE that avoid error propagation (standard-compliant PE), and PE that try to exploit that property for security. In both cases we show that attacks that require complexity much lower than exhaustive enumeration of encrypted/key bits can successfully yield good quality content. Experimental results are shown for images, but the conclusions can be extended to partial encryption of video and other types of media.

1. INTRODUCTION

Encryption is a fundamental technology for avoiding piracy of multimedia content. Strong encryption methods, like triple-DES [2] and AES [4], can provide security for the most demanding civil applications, but their computational cost is considered too high for consumer applications, like high-definition digital video. At the same time, recent events have shown that software to break unsafe systems can be rapidly disseminated in the Internet, and strong security is essential even in home applications. One way to achieve inexpensive security is to use the fact that the complexity and cost of protecting multimedia files can be greatly reduced by using partial encryption (PE), which is the process of encrypting only a carefully selected fraction of the bits in the compressed data.

An interesting type of PE preserves the syntax of the compressed data. The advantage of this approach is that

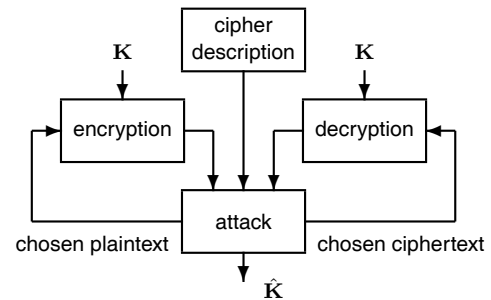


Fig. 1. Model for cryptanalysis of general-purpose ciphers.

any standard-compliant decoder can, without the key, still decompress and render a version of the media, with quality severely degraded, but still discernible. This can be an attractive feature for commercial multimedia, when it is desirable to let customers try a low-quality version before buying. At the same time, piracy is avoided if it is not feasible to obtain the high-quality version without the correct key. One example is the encryption of only the DCT AC coefficient signs in JPEG and MPEG files [7].

Another form of PE exploits the fact that some bits of the compressed data affect all the variable-length coding after it. When these bits are encrypted, it becomes impossible to parse and recognize the meaning of all the subsequent non-encrypted bits. One example is the partial encryption of significance data for SPIHT image compression [5].

In this work we provide evidence that both strategies can significantly compromise security by showing that even low-complexity attacks allow retrieval of significant amounts of encrypted data without the knowledge of the encryption key. We discuss the problem of measuring the strength of PE, and propose a model based on the complexity to recover the media to a certain distortion—instead of the complexity to recover the encryption key—using side information.

2. PROPOSED CRYPTANALYSIS MODEL

The security of partial encryption is frequently assumed to be similar to other private-key encryption, and that consequently its strength should also be measured solely by the

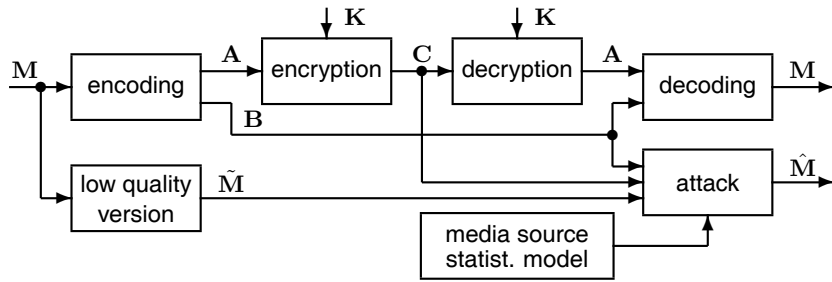


Fig. 2. Proposed model for cryptanalysis of multimedia ciphers: the success is measured by the difference in distortion $D(M, \tilde{M}) - D(M, \hat{M})$.

difficulty of finding the encryption key. Fig. 1 shows such a model for general-purpose ciphers, which enables evaluating the complexity of finding the key under attacks like known-plaintext, chosen-plaintext, etc [1, 2]. It is also commonly assumed that, if the strength is not defined by the complexity to find the key, then it can only be measured by the complexity to recover all the encrypted data.

Unfortunately, there are several problems with these approaches for evaluating strength.

1. It does not take into account exploiting information in the unencrypted data. It is necessary to consider the fact that not all redundancy is eliminated by compression, and thus it may be practical to guess the encrypted bits using the unencrypted bits.
2. An assumption like known-plaintext is not useful if the objective is not to find the key. On the other hand, multimedia data has very important and well known statistical and structural properties, which should be a part of the attack model, replacing the known-plaintext assumption.
3. Commercial multimedia, by its own nature, cannot avoid the access of important information by attackers. In fact, the availability of low-quality versions, obtained from analog signals, demo versions, etc., is more the rule than the exception.

We believe that the model shown in Fig. 2 provides a more useful methodology for evaluating the security of partial encryption on commercial applications. We assume that a low quality version of the media is somehow known to the attacker, and that the effectiveness of the attack is measured by the complexity to reduce its distortion (improve its subjective quality). Fig. 3 shows complexity vs. distortion curves for hypothetical encryption methods and attacks. We can see that in the ideal case it is not possible to reduce distortion unless we retrieve the encryption key with exhaustive enumeration. A slightly weaker system can yield improvements but only at great costs. Weak systems, on the

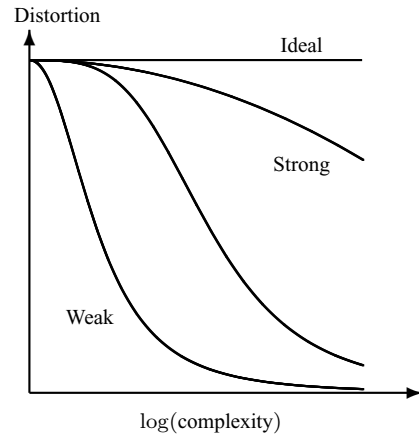


Fig. 3. Examples of distortion-complexity curves of different partial-encryption methods.

other hand, permit quick improvement with low-complexity attacks.

This analysis shows the inappropriateness of using the zero-distortion criterion to measure the encryption strength: even a weak system may require complexity higher or equivalent to finding the key by exhaustive enumeration to reach zero distortion.

3. EXAMPLES OF LOW-COMPLEXITY ATTACKS

3.1. Encryption of DCT Coefficient Signs

Let us consider a simple form of partial encryption for JPEG images in which only the signs of the DCT AC coefficients are encrypted. This creates standard-compliant files because the signs use only one bit, and do not affect the meaning of other bits. Thus, encrypted images can still be seen on any standard viewer. Many variants of this method, including the encryption of the DC-difference sign, and other data, had been proposed for JPEG and MPEG files (e.g., [7]).

If we use a strong cipher like AES to encrypt those bits, the complexity can still be low because signs represent a small fraction of the total number of bits, and we can be

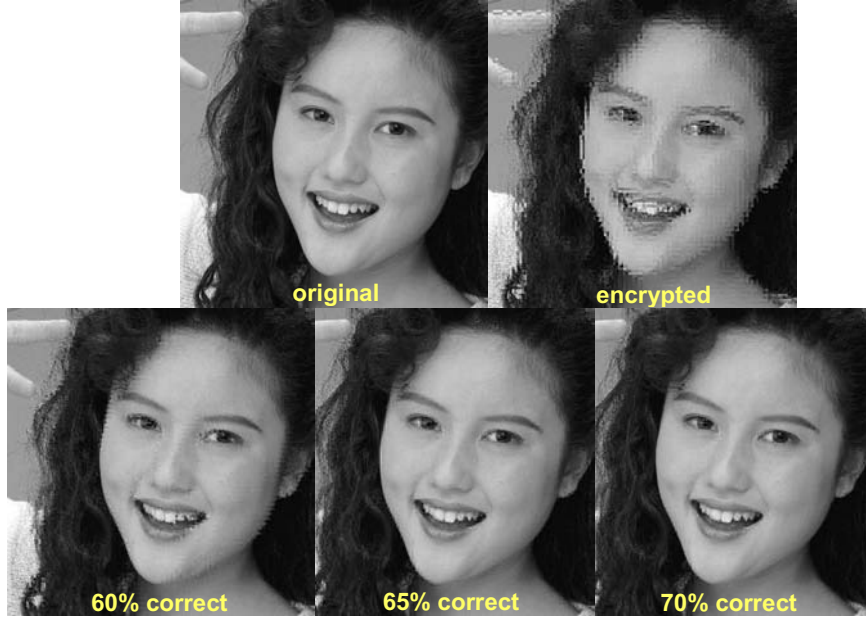


Fig. 4. Results of attack on DCT-sign partial encryption.

sure that it is currently impractical to try to recover the key. However, we can try to guess these signs from side information (e.g., thumbnail image) and from the unencrypted bits.

We considered the fact that the creation of format-compliant files can create a vulnerability because the *meaning* of all bits in the value are known, only the *value* of the signs is unknown. Thus, we can devise attacks based on multiple passes on the image file, each time guessing a subset of encrypted bits from the rest of the data. For instance, we can first guess the signs in the 8×8 blocks with the smallest number of AC coefficients (unencrypted information).

The statistical properties do not have to be explicitly stated for estimating the signs. We use the fact that a significant amount of knowledge is available about image interpolation and restoration. Let $\hat{\mathbf{b}}_{i,j}$ be the current best estimate of the 8×8 (i, j) block. We can represent the image block interpolation by function $\Gamma(\cdot)$

$$\tilde{\mathbf{b}}_{i,j} = \Gamma(\hat{\mathbf{b}}_{i-1,j-1}, \hat{\mathbf{b}}_{i-1,j}, \hat{\mathbf{b}}_{i-1,j+1}, \hat{\mathbf{b}}_{i,j-1}, \hat{\mathbf{b}}_{i,j+1}, \hat{\mathbf{b}}_{i+1,j-1}, \hat{\mathbf{b}}_{i+1,j}, \hat{\mathbf{b}}_{i+1,j+1}).$$

defining the interpolated block $\tilde{\mathbf{b}}_{i,j}$.

We estimate the new vector with signs $\mathbf{s}_{i,j}$ using

$$\mathbf{s}_{i,j} = \arg \min_{\mathbf{s}_{i,j}} |\tilde{\mathbf{b}}_{i,j} - \mathbf{T}_{i,j} \mathbf{s}_{i,j}|, \quad (1)$$

where $\mathbf{T}_{i,j}$ is the matrix corresponding to the inverse DCT with the AC coefficient magnitudes specified in the partially-encrypted file. The new estimate is then

$$\hat{\mathbf{b}}_{i,j} = \mathbf{T}_{i,j} \mathbf{s}_{i,j}.$$

It may seem that, because the elements of vector $\mathbf{s}_{i,j}$ belong to the discrete set $\{-1, 1\}$, the minimization in eq. (1) would have complexity $O(2^{N_{i,j}})$, where $N_{i,j}$ is the number of signs in the block. However, because the DCT is an orthogonal transformation, the problem is completely separable, and the complexity is actually $O(N_{i,j})$ per block. Consequently, the complexity of this attack is $O(\sum_{i,j} N_{i,j})$, which is extremely low for a cryptographic application.

The effectiveness of this attack depends on how good is the interpolation. Fig. 4 show the results of such attack on a compressed JPEG image, using different interpolation methods, and assuming recovered neighbor blocks. In the first case, using simple linear interpolation and a small number of passes produced an image in which the number of correct sign bits increased from 50% to 60%. The second and third cases, use more complex interpolation based on linear transforms and thresholding [6], yield improvements of 65% and 70%. We can see that, even though a significant fraction of the encrypted bits could not be recovered, the attack successfully improved the quality with relatively low complexity.

3.2. Encryption of Significance Data

Since in the previous section we exploit the independence of the encrypted bits for a successful attack, it may seem that increasing dependence will always improve security. However, this is not necessarily true. Another scheme [5] for partial encryption proposes the encryption of significance data in the SPIHT [3] image/video compression. Since the mean-



Fig. 5. Results of attack on partial encryption of SPIHT significance.

ing of the compressed bits depends on the order in which set/pixel data is added and removed from lists, encrypting the significance makes it very hard to know the actual meaning of the remaining bits, and correct decoding.

At the same time, this type of “error propagation” can be used for error detection. Suppose we have some low-quality version of the image $\tilde{M}$, and let v_0 and v_1 be vectors with N bits, always starting with a 0-bit and 1-bit, respectively, that are going to be used to replace the encrypted significance bits. Let $\hat{M}(v_0)$ and $\hat{M}(v_1)$ be the corresponding images obtained by decoding only up to the N -th encrypted significance bit using these vectors. With the full enumeration of all possible v_0 and v_1 , we can compute

$$D_b = \min_{v_b} |\hat{M}(v_b) - \tilde{M}|, \quad b = 0, 1.$$

Since an incorrect bit inevitably leads to random image parts, and the correct bits leads to a proper image, if N is sufficiently large, when we test all vectors we find that the correct first bit b is such that $D_b \ll D_{1-b}$. Consequently, an attack can be implemented by searching a sufficiently large number of bit vectors, and deciding on each bit sequentially. Since $\tilde{M} \neq M$, the attack must eventually stop when the distance between images cease to decrease.

Fig. 5 shows the application of such attack to the case in which the original image is partially encrypted, and the attacker has a thumbnail image with a visual watermark (Fig. 5a). We use settings similar to [5], encrypting about 2% of the bits. Remarkably, the majority of the first 480 encrypted bits can be found testing less than 2000 brute-force tests, while two bits required more than 2^{20} tests, and five bits required more than 2^{24} tests. The partially image decoded image was used to try to remove the watermark, and the result is shown in Fig. 5b. Applying the attack to a reduced number of encrypted bits requires much less effort, and the result shown in Fig. 5c demonstrates that with much larger search time it is possible to keep removing the watermark, and also adding high-frequency components (missing from the thumbnail).

4. CONCLUSIONS

We consider how partial encryption techniques can compromise security by proposing a new model for attacks and for measuring security, based on the fact that for multimedia low quality versions are commonly available (e.g., from analog signals, free demo versions). In this model the objective is not to recover the key, but to use the partially encrypted data to improve the quality of the known version. While these are preliminary theoretical results, the usefulness of the new model is shown using two new attacks on two types of partially encrypted compressed images. Even though we could not recover the key, nor the exact original, our results show that, using the information contained in the non-encrypted data, significant improvements in quality could be obtained with relatively low complexity.

5. REFERENCES

- [1] D.R. Stinson, *Cryptography Theory and Practice*, CRC Press, Boca Raton, 1995.
- [2] B. Schneier, *Applied Cryptography*, 2nd ed., John Wiley & Sons, Inc., New York, 1996.
- [3] A. Said and W.A. Pearlman, “A new fast and efficient image codec based on set partitioning in hierarchical trees,” *IEEE Trans. Circ. Syst. Video Tech.*, vol. 6, pp. 243–250, June 1996.
- [4] National Institute of Standards and Technology (NIST), *Advanced Encryption Standard (AES)*, FIPS Pub. 197, U.S. Dept. of Commerce, Nov. 2001.
- [5] H. Cheng and X. Li, “Partial encryption of compressed images and video,” *IEEE Trans. on Signal Processing*, vol. 48(8), pp. 2439–2451, Aug. 2000.
- [6] O.G. Guleryuz, “Nonlinear approximation based image recovery using adaptive sparse reconstructions,” *Proc. IEEE Int. Conf. Image Processing*, vol. 1, pp. 14–17, Sept. 2003.
- [7] W. Zeng and S. Lei, “Efficient frequency domain selective scrambling of digital video,” *IEEE Trans. Multimedia*, vol. 5(1), pp. 118–129, March 2003.